

Non-radioactive Phosphorylation with T4 PNK or T4 PNK (3' phosphatase minus)

Overview

Protocols.io also provides an interactive version of this protocol where you can discover and share optimizations with the research community. It is available for both the [Non-Radioactive Phosphorylation with T4 PNK](#) and the [Non-Radioactive Phosphorylation with T4 PNK \(3' phosphatase minus\)](#).

1. Set-up the following reaction in a microcentrifuge tube on ice:

DNA	up to 300 pmol of 5' termini
T4 PNK Reaction Buffer (10X)	5 µl
ATP (10 mM)	5 µl
T4 PNK	1 µl (10 units)
Nuclease-free Water	up to 50 µl

2. Incubate at 37°C for 30 minutes.

3. Heat inactivate by incubating at 65°C for 20 minutes.

Notes:

- ATP is not supplied
- 1X T4 DNA Ligase Buffer contains 1 mM ATP and can be substituted in non-radioactive phosphorylations (T4 Polynucleotide Kinase exhibits 100% activity in this buffer).